Unit-Nets Day 3:
Transplants, Integer Rows, and the Highways of Syntax

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Third in the series (Day 1: *One Day of Training Without Gradients*; Day 2: *A Language Model You Can Read With Arithmetic*). Three ideas from Remy; all three ran to results; one rewrote the migration strategy.

1 Idea 1 — weight transplant instead of distillation

Make pretrained weights legal *in place*: normalize each row's positives to sum 1, bound negatives by the row's most negative magnitude.

Naive: total collapse. One constrained Qwen block → 0.4% agreement; all 24 → 0.0%. Row normalization destroys per-row output scale (positive sums run 10–100); transformers cannot absorb it downstream — Day 1's post-hoc-projection lesson at 500M scale.

Folded: nearly free. Use one per-row divisor (the positive sum) and fold it exactly into the next matrix wherever the connecting path is linear: `v_proj`→`o_proj` (attention mixes values linearly; needs GQA-aware scale replication and scaling v's *bias*) and `up_proj`→`down_proj` (SwiGLU is linear in `up`). Blocked, each for a nameable reason: q/k (RoPE rotates dimension pairs), gate (SiLU inhomogeneous), o/down (residual stream — nowhere left to push the scale debt).

Then the ResNet: CNNs are built from exactly the two mechanisms that make folding total — BatchNorm absorbs scale (μ/s, σ^2/s^2, exact ϵ-correction in γ) and ReLU is positively homogeneous.

	agreement	clipped negatives	weights legal
Qwen naive	0.000	—	—
Qwen folded (v+up)	0.977	0 / 107M	30.4%
ResNet18 folded (all conv-BN)	**0.9995**	**0 / 11.2M**	**95.5%**

Imagenette top-1 identical (65.32% vs 65.30%); only the fc head (no BN behind it) resists. **A pretrained ResNet is a unit-net that doesn't know it yet.** Transplant (97.7–99.95%) beats distillation (33%) so thoroughly that the migration path is now: *transplant what folds, constrained-finetune the rest.*

2 Idea 2 — native integer quantization

A convex P row stored as integer numerators summing to exactly 2^b (largest remainder) is exact fixed-point arithmetic with a shared denominator; N takes the 2^b grid on $[0, 1]$.

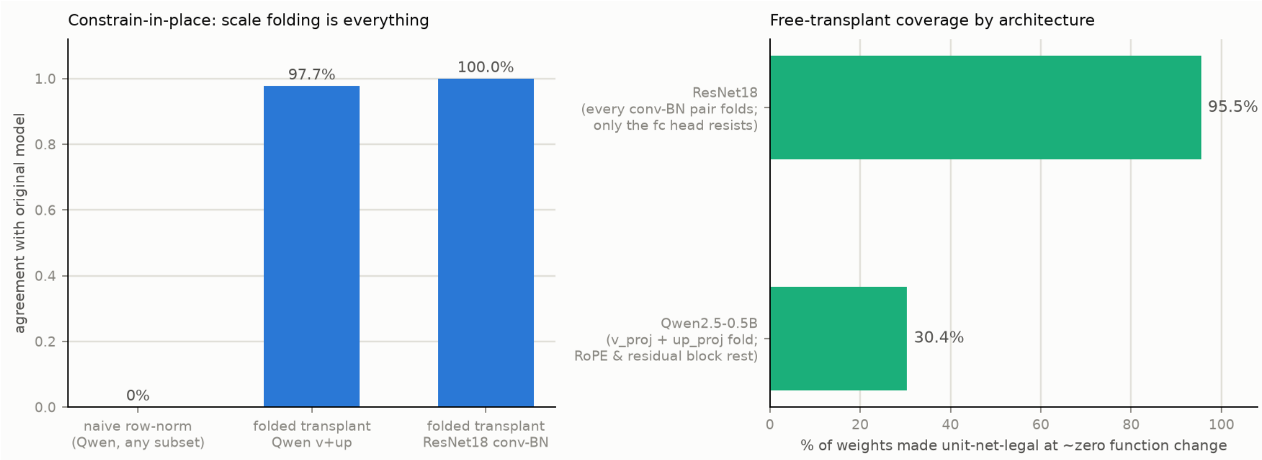

Figure 1: Constrain-in-place: folding is everything; coverage by architecture.

bits	LM agree	LM pos-slot	ResNet top-1
float32	0.300	1.00	0.653
12	0.302	1.00	0.653
10	**0.300 (lossless)**	1.00	0.410
8	0.214	1.00	0.000
4	0.096	1.00	0.000
2	0.001	1.00	0.000

For the LM student, 10 bits is lossless *and* 5× sparsifying (3.4M → 710k nonzeros): the integer budget forces zeros, so the bit budget *is* a sparsity budget. The same coupling is a liability for lean deep nets: ResNet's 4,608-weight filters amputate to ≤256 nonzeros at 8 bits and 20 layers compound the noise. **Law: simplex quantization needs denominator ≥ row length; tolerance to constraint-native compression measures the redundancy training left behind.**

3 Idea 3 — activation-flow pruning

Importance = corpus-summed activation flow per edge ($\mathbb{E}[|w| \cdot a]$); raise the floor, watch divergence. Pruned models stay legal (P rows renormalized; for the ResNet the fresh scale re-folds into BN, so only the pruning itself changes the function).

edges pruned	LM agree	LM KL	LM pos-slot	ResNet top-1
50%	0.239	0.81	1.00	0.524
80%	0.094	3.77	1.00	0.103
90%	0.024	7.69	1.00	0.000
99%	0.000	15.3	1.00	—

4 The cross-cutting finding: syntax rides the highways

Across every destructive condition on the LM — 2-bit quantization, 99% pruning, agreement statistically zero — the pos-slot syntax task held **1.00**. The verb/noun circuit lives on the highest-flow edges, which every mass-respecting destruction spares last; semantic capillaries die first. With Day 2's j-carve: targeted carving keeps the circuit at 4% of the network *on purpose*; flow-pruning keeps it at 1% *by accident*. *Syntax rides the trunk roads; meaning lives in the side streets.*

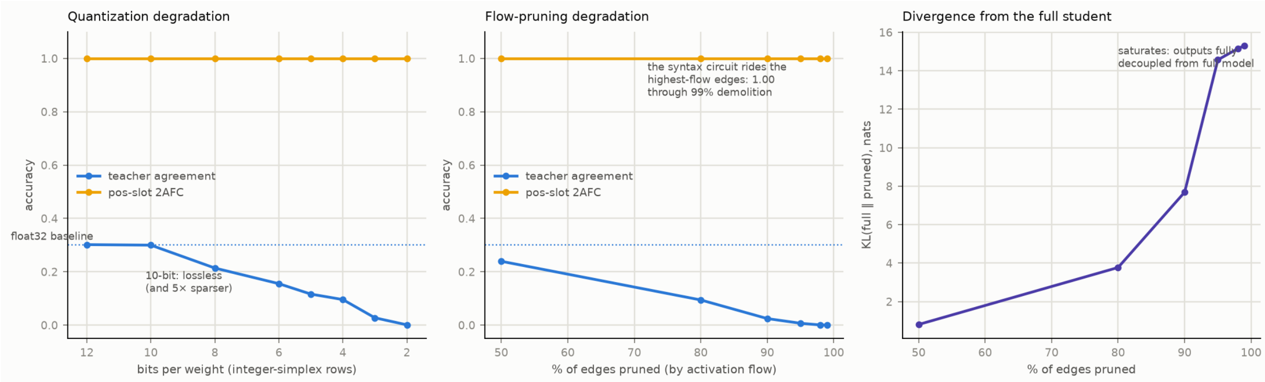

Figure 2: LM student: quantization and flow-pruning degradation; divergence from the full model. The yellow line is §4.

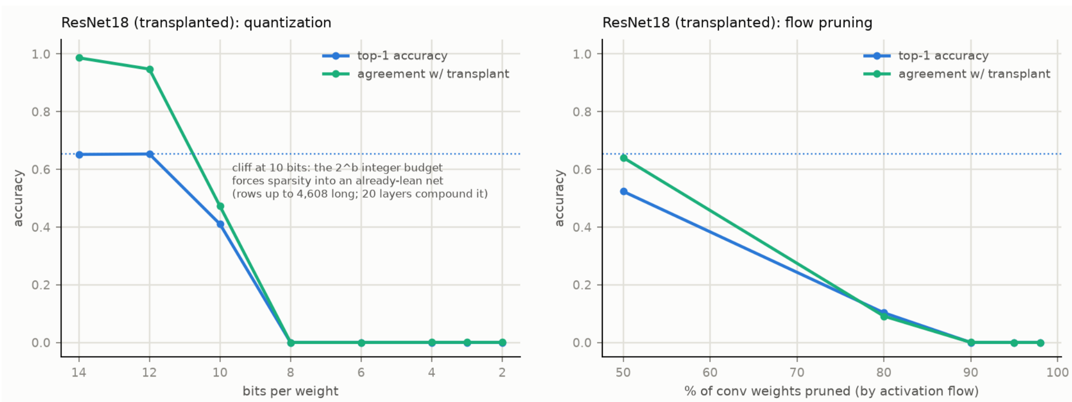

Figure 3: ResNet18: the same treatments; steeper everywhere — depth compounds and lean filters amputate.

5 Artifacts

Channel-level 3D explorer (3,917 channel-nodes, 9,620 edges, residual arcs; conv1 nodes open their actual RGB 7×7 filters — each now a unit budget, so intensities compare across the layer): artifact c51a40dc. Voxel-level whole-net explorer (19 stages, 1,098 voxels lit by real activations on a church image, 13,784 exact receptive-field cones as low-clearance curves, orbit+pan+zoom; weight sharing directly visible): artifact ff2ee392. (v1's lofted arcs cached for posterity.)

6 Standing conclusions

(1) Constraint migration is mostly free where normalization absorbs scale: ∼100% of CNN convs, ∼30% of transformer linears, exactly, with zero clipping observed. (2) The geometry quantizes and sparsifies as one operation, with a sharp validity condition. (3) Robustness to native compression is a redundancy meter. (4) Functional anatomy is stratified by flow: syntax on highways, semantics on capillaries — measured, not metaphor.

Next: constrained-finetune the transplants to retire the scale debt; rerun the J-lens/workspace program on the transplanted ResNet (95.5% legal = the exact lens now applies to a real ImageNet model); j-carve Tier-1 against a transplant-initialized student.